

Motion-Controlled Game

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Date: September 2024

1 Objective

Develop an interactive game where the digital robot arm simulate players' movement in real-time, by using IMU and EMG detectors.

2 User Guide

2.1 software requirement

To set up the development environment, you'll need to install Python and a few other tools. Here's a step-by-step guide:

1. Python Installation:

- You'll need to download and install Python, version 3.12 or higher. Visit the official Python website at <https://www.python.org/>. Search for the latest version, and download the one that meets or exceeds version 3.12.

2. Installing Required Python Packages:

- Once Python is installed, you'll need to install a few essential packages. Open your command line or terminal and run the following command:

```
pip install matplotlib numpy pyserial pillow
```

3. Unity Installation:

- The game is built on the Unity platform, so you'll need to download Unity. Before proceeding, make sure your system meets the configuration requirements specified in Appendix A.
- Head over to the Unity download page at <https://unity.com/download>. Follow the instructions provided on the site to download two applications: Unity and Unity Hub.
- After installing Unity Hub, open it, click "Add," and select the Unity package.

4. Blender Installation (if needed):

- The game model is created using Blender. If you encounter any errors related to Blender's environment, you'll need to install it as well.
- Check the requirements mentioned in Appendix B to ensure your system is compatible.
- Visit the Blender download page at <https://www.blender.org/download/> and follow the instructions to download and install the software.

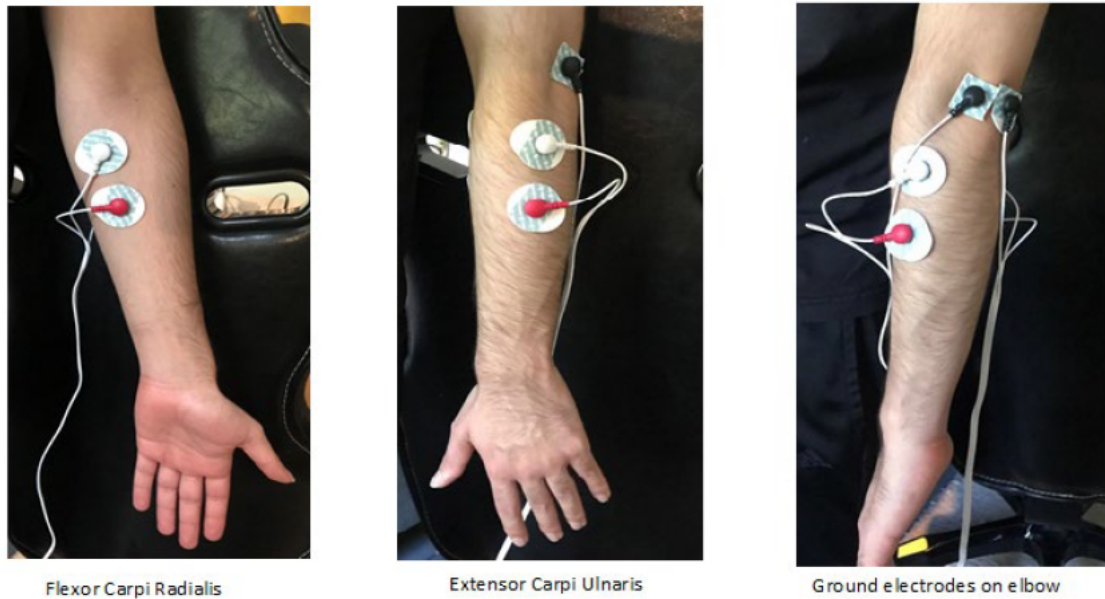
2.2 sleeve

The sleeve is equipped with two detectors. The first one is an IMU (Inertial Measurement Unit) that measures the movement of the arm. This detector should be positioned on the front, centered on the wrist, like this:



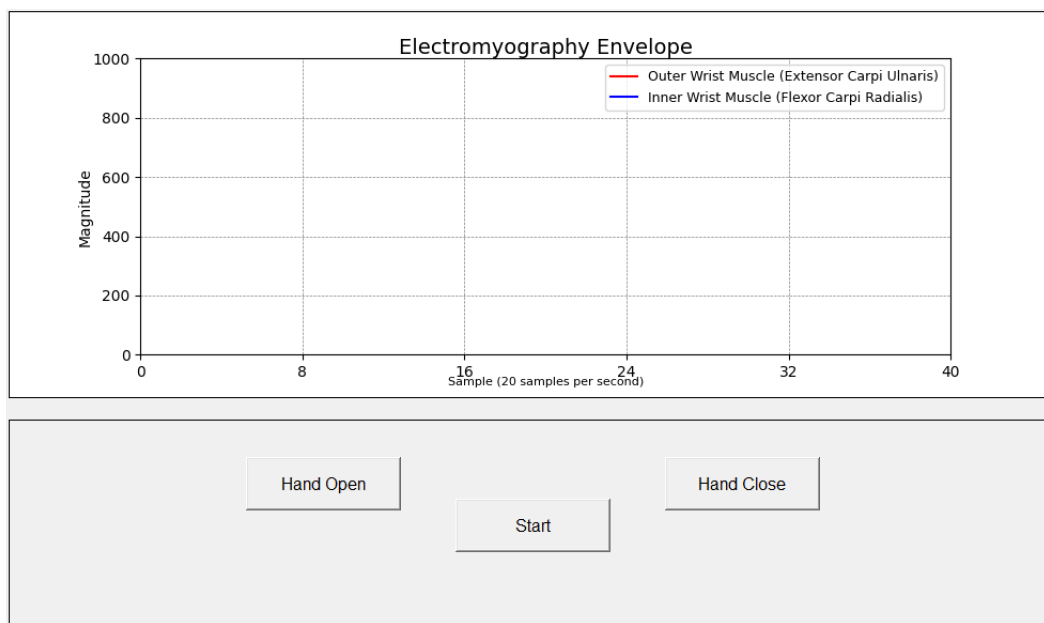
The second detector is an EMG (Electromyography) sensor, which is connected to six electrodes. These measure signals from two main muscles. There's an instruction on how to set it up.

You will need to stick the electrodes in the positions as shown in the figure below:



2.3 game preparation

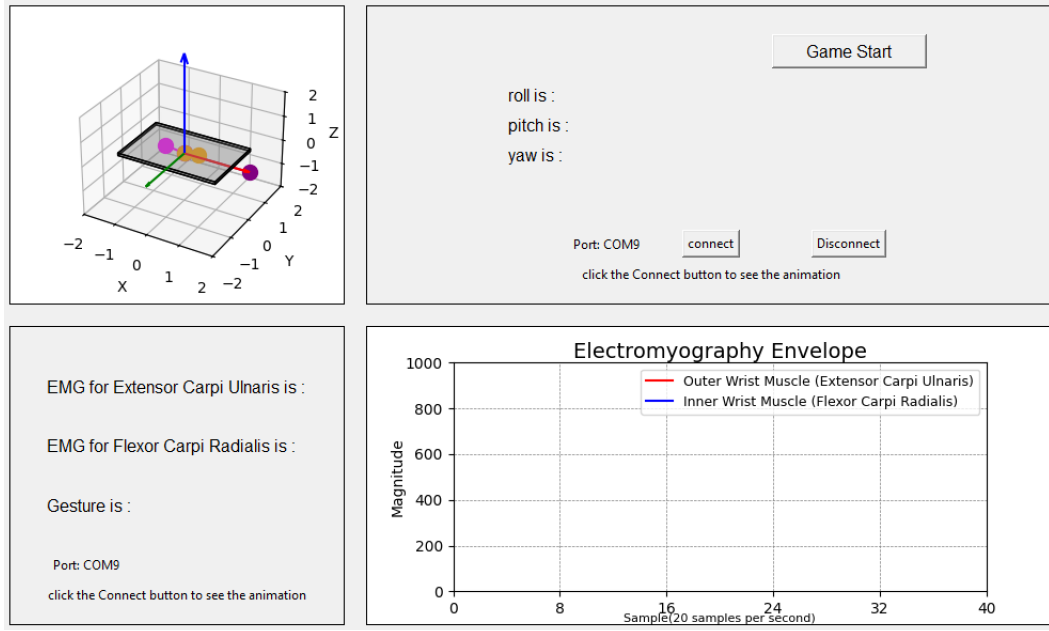
After putting on the sleeve, open the software (the Python file: Window2). You'll see the cover screen—click the "Start" button to enter the training section.



Click the "Hand Open" button to record data with your hand in the open position. Similarly, click the "Hand Close" button to record data with your hand in the closed position.

After the training session, the next interface will display how the data is processed, showing

graphs, analysis, and predictions based on the recorded data.



After reviewing the analysis and predictions, click the "Disconnect" button. Then, open the game in Unity and click on the "Game Start" button. The analyzed data will be transferred to Unity, and the game will begin.

2.4 rule instruction

In the game, on the table, there are three moving balls, but only one is the correct one. Find the correct ball, pick it up, and place it in the basket. In the limited time, the people who take most balls to the basket will be the winner.

3 Data analysis

3.1 IMU data analysis

For IMU data analysis, the IMU detector have an output of Quaternion numbers and the acceleration, Use the equations below we can calculate the raw, roll, pitch of the movement.

```
roll = math.atan2(2 * (q0 * q1 + q2 * q3), 1 - 2 * (q1 * q1 + q2 * q2))
```

```
pitch = -math.asin(2 * (q0 * q2 - q3 * q1))
```

```
yaw = -math.atan2(2 * (q0 * q3 + q1 * q2), 1 - 2 * (q2 * q2 + q3 * q3))
```

In addition,after finding the difference of these real-time output, we can know how much angle a joint turn and send to the unity then the digital robot will have the respond movement.

3.2 EMG data analysis

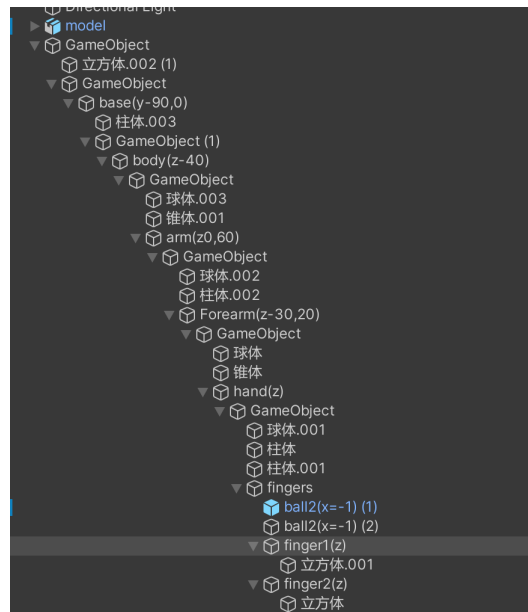
For the EMG data analysis, I have two alternative methods. The first method involves extracting features from the data for different gestures and then plotting a graph. In this graph, lines will be used to divide the data into different labeled regions. During the prediction phase, we can simply identify which region the data point falls into, and the corresponding label will indicate the real-time gesture.

The second method involves using machine learning. I use a Multi-Layer Perceptron (MLP) to train a model. The model will output a labeled point that indicates which gesture is being performed.

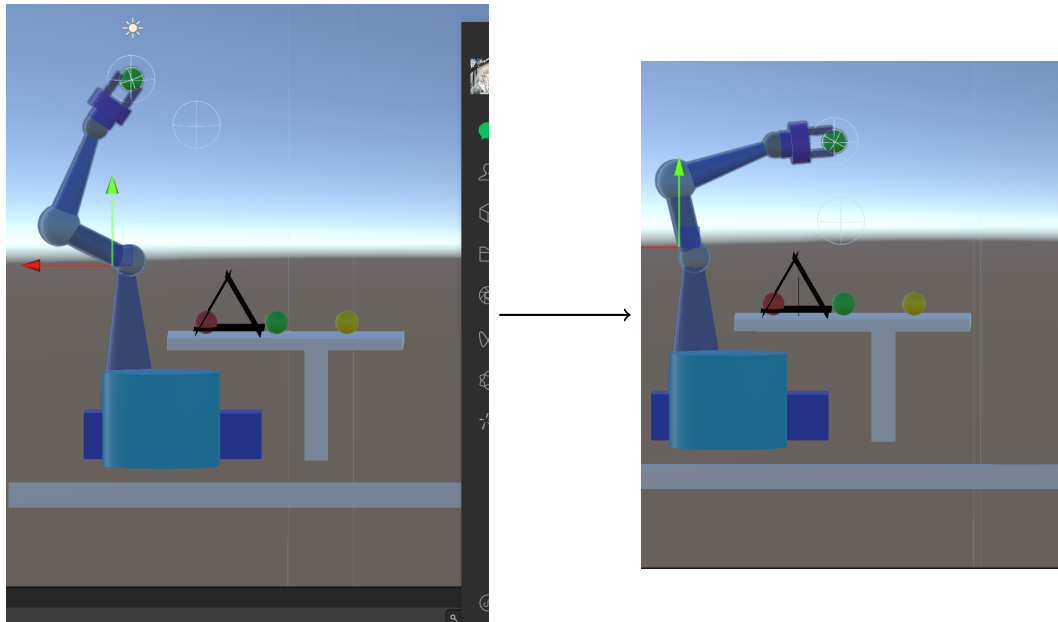
Both methods have their own advantages and disadvantages. The first method saves time in model development, but it requires that the features be relatively simple and distinct to ensure the graph can be easily divided.

4 Building the digital arm in the Unity

I created a digital arm model in Blender and imported it into Unity. In this setup, each part of the arm is separated, meaning the movement of one joint does not affect the others. To solve this problem I use the Hierarchy (shown in picture below), which can set up a parent-child relationship between the joints, we can achieve the effect where moving the upper arm also moves the forearm accordingly. For example, by making the forearm a child object of the upper arm, any movement of the upper arm will automatically cause the forearm to follow. This hierarchical structure ensures that the movement of each joint influences the others, resulting in more realistic limb motion.



In addition, I created a single empty object as the center for each joint, positioned at the end of the joint. This setup ensures that the rotation of each joint is centered around its end, allowing for more natural and accurate joint movements, we can see the example picture below. In the examples, it demonstrates that when one joint rotates by 50 degrees around the Z-axis, the entire arm will also rotate by 50 degrees.



5 Conclusion and further studies

- I have calculated the roll,yaw and pitch, further study can calculate the displacement and even the real time location.
- In the unity, how to show the displacement is also need to considered, Originally, I mentioned that the center of each joint is at the end of the joint. Instead, we can consider setting a center point for the entire arm, located in the position of the hand, and move the whole arm based on this center point's displacement or the real-time location.
- For the development on EMG data analysis, Originally, we only have two gestures, in the further studies we can have move gestures, and have more labels when building the MLP label.
- Each EMG sensor requires six electrodes, which can be time-consuming to put on. To address this, we should consider designing a more convenient glove or alternative type of sensor that reduces the time needed for setup.

Appendix

A Unity Editor Requirement

Unity Editor system requirements

This section lists the minimum requirements to run the Unity Editor. Actual performance and rendering quality may vary depending on the complexity of your project.

Minimum requirements	Windows	macOS	Linux (Support in Preview)
Operating system version	Windows 7 (SP1+), Windows 10 and Windows 11, 64-bit versions only.	High Sierra 10.13+	Ubuntu 20.04, Ubuntu 18.04, and CentOS 7
CPU	X64 architecture with SSE2 instruction set support	X64 architecture with SSE2 instruction set support	X64 architecture with SSE2 instruction set support
Graphics API	DX10, DX11, and DX12-capable GPUs	Metal-capable Intel and AMD GPUs	OpenGL 3.2+ or Vulkan-capable, Nvidia and AMD GPUs.
Additional requirements	Hardware vendor officially supported drivers	Apple officially supported drivers	Gnome desktop environment running on top of X11 windowing system, Nvidia official proprietary graphics driver or AMD Mesa graphics driver. Other configuration and user environment as provided stock with the supported distribution (Kernel, Compositor, etc.)
	For all operating systems, the Unity Editor is supported on workstations or laptop form factors, running without emulation, container or compatibility layer.		

B Blender Requirement

Windows

	Minimum	Recommended
OS	Windows 8.1 (64-bit)	Windows 10 or Windows 11
CPU	4 cores with SSE4.2 support	8 cores
RAM	8 GB	32 GB
GPU	2 GB VRAM with OpenGL 4.3 (see below)	8 GB VRAM

NVIDIA: [GeForce](#) 400 and newer, Quadro Tesla GPU architecture and newer, including [RTX-based](#) cards, with NVIDIA drivers.

AMD: [GCN 1st gen](#) and newer.

Intel: [Broadwell](#) architecture and newer. Always make sure to install the latest drivers from the graphics card manufacturer website.

macOS

	Minimum	Recommended
OS	macOS 11.2 (Big Sur)	macOS 14 (Sonoma)
CPU	Apple Silicon or Intel	Apple Silicon
RAM	8 GB	32 GB
GPU	GPU with Metal 2.2 (see below)	

AMD, Apple Silicon, or Intel (Skylake and newer).

Linux

	Minimum	Recommended
OS	Distribution with glibc 2.28 or newer (64-bit)	
CPU	4 cores with SSE4.2 support	8 cores
RAM	8 GB	32 GB
GPU	2 GB VRAM with OpenGL 4.3 (see below)	8 GB VRAM

NVIDIA: [GeForce](#) 400 and newer, Quadro Tesla GPU architecture and newer, including [RTX-based](#) cards, with NVIDIA drivers.

AMD: [GCN 1st gen](#) and newer.

Intel: [Broadwell](#) architecture and newer. Always make sure to install the latest drivers from the graphics card manufacturer website.